

Intensity and timing of physical activity in relation to postmenopausal breast cancer risk: the prospective NIH-AARP Diet and Health Study



Background Despite strong evidence of an inverse association of physical activity with postmenopausal breast cancer risk, whether a certain intensity or time of life of physical activity is most effective for lowering breast cancer risk is not known. **Methods** In 118,899 postmenopausal women in the prospective NIH-AARP Diet and Health Study, we examined the relations of light and moderate-to-vigorous intensity physical activity during four periods of life ("historical": ages 15-18, 19-29, 35-39 years; "recent": past 10 years) to postmenopausal breast cancer risk. Physical activity was assessed by self-report at baseline, and 4287 incident breast cancers were identified over 6.6 years of follow-up. **Results** In age-adjusted and multivariate Cox regression models, >7 hours/week of moderate-to-vigorous activity during the past 10 years was associated with 16% reduced risk of postmenopausal breast cancer (RR:0.84; 95%CI:0.76,0.93) compared with inactivity. The association remained statistically significant after adjustment for BMI (RR:0.87; 95%CI:0.78,0.96). Neither moderate-to-vigorous activity during other periods of life nor light intensity activity during any period of life was related to breast cancer risk, and associations did not vary by tumor characteristics. **Conclusion** A high level of recent, but not historical, physical activity of moderate-to-vigorous intensity is associated with reduced postmenopausal breast cancer risk. More precise recall of recent physical activity than activity in the distant past is one possible explanation for our findings.